Hawking could see, in his congealed mental pictures and diagrams, that the areas of absolute horizons (but not necessarily apparent horizons) will increase not only when black holes collide and coalesce, but also when they are being born, when matter or gravitational waves fall into them, when the gravity of other objects in the Universe raises tides on them, and when rotational energy is being extracted from the swirl of space just outside their horizons. Indeed, the areas of absolute horizons will almost *always* increase, and can never decrease. The physical reason is simple: Everything that a hole encounters sends energy inward through its absolute horizon, and there is no way that any energy can come back out. Since all forms of energy produce gravity, this means that the hole's gravity is continually being strengthened, and correspondingly, its surface area is continually growing.

Hawking's conclusion, stated more precisely, was this: *In any region of space, and at any moment of time (as measured in anyone's reference frame), measure the areas of all the absolute horizons of all black holes and add the areas together to get a total area. Then wait however long you might wish, and again measure the areas of all the absolute horizons and add them If no black holes have moved out through the "walls" of your region of space between the measurements, then the total horizon area cannot have decreased, and it almost always will have increased, at least a little bit*

Hawking was well aware that the choice of definition of horizon, absolute or apparent, could not influence in any way any predictions for the outcomes of experiments that humans or other beings might perform; for example, it could not influence predictions of the waveforms of gravitational radiation produced in black-hole collisions (Chapter 10), nor could it influence predictions of the number of X-rays emitted by hot gas falling into and through a black hole's horizon (Chapter 8). However, the choice of definition could strongly influence the *ease* with which theoretical physicists deduce, from Einstein's general relativistic equations, the properties and behaviors of black holes. The chosen definition would become a central tool in the paradigm by which theorists guide their research; it would influence their mental pictures, their diagrams, the words they say when communicating with each other, and their intuitive leaps of